9.Implementation of a Basic Reinforcement Learning Model (Q-Learning)

Aim :

To implement the **Q-Learning** algorithm, a model-free reinforcement learning technique in which an **agent** learns an optimal policy by interacting with an **environment** and updating a **Q-table** of state-action values, and to analyze its performance using **cumulative reward per episode, steps per episode, convergence of the Q-values** and the **learned policy**.

Algorithm :

A. Q-Learning Implementation

Input: Environment (states **S**, actions **A**, rewards **R**); learning rate **α**; discount factor **γ**; exploration rate **ε**; number of episodes **N**

Step 1: Initialise the Q-table **Q(s, a) = 0** for all states s ∈ S and actions a ∈ A.

Step 2: For **each episode = 1 to N**:

1. Reset the environment and obtain the initial state **s**.
2. Repeat for each step until the **terminal state** (or maximum steps) is reached:

• **Choose an action (ε-greedy policy):** with probability **ε** select a random action (**exploration**); otherwise select **a = argmax Q(s, a)** (**exploitation**).

• **Take action a**, then observe the reward **r** and the next state **s'**.

• **Update the Q-value (Bellman equation):**

$Q(s, a) \leftarrow Q(s, a) + \alpha \cdot [\, r + \gamma \cdot \max Q(s', a') - Q(s, a) \,]$

• **Move to the next state:** s ← s'.

3. **Decay ε** after each episode (for example, ε ← max(εmin, ε × decay)) to shift from exploration to exploitation.
4. Record the **total reward** and the **number of steps** of the episode.

Step 3: Output the learned Q-table and the optimal policy:

$\pi(s) = \text{argmax } Q(s, a)$

B. Performance Analysis Procedure

Step 1: Define the **environment** (for example, a Grid World or the FrozenLake environment)
with its states, actions and rewards.

Step 2: Set the **hyperparameters** (α, γ, ε, ε decay, number of episodes).

Step 3: Train the agent using the Q-Learning algorithm.

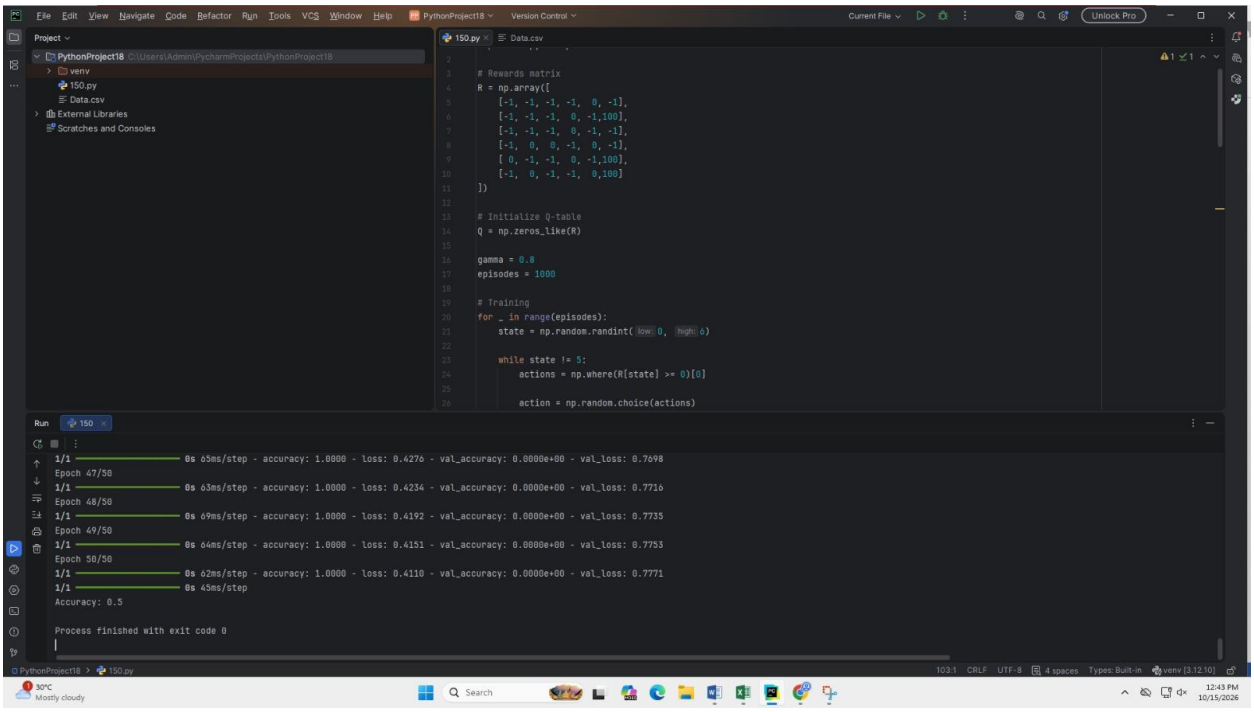
Step 4: Plot the **total reward per episode** and the **steps per episode** to observe learning progress.

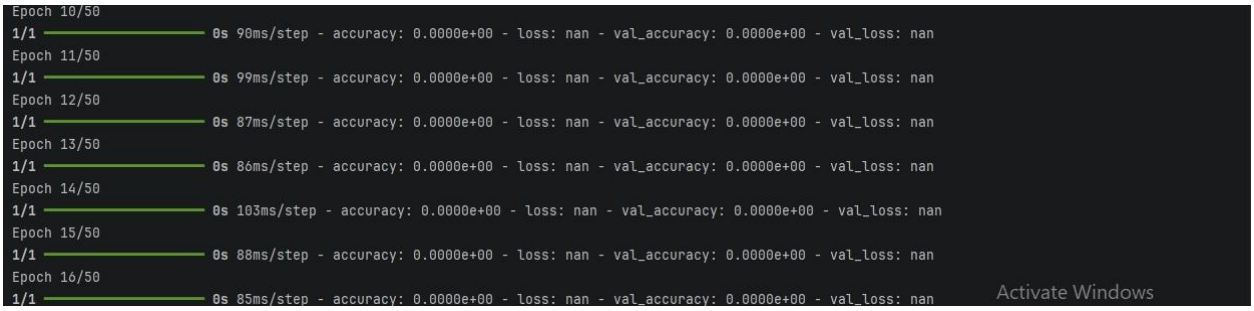
Step 5: Display the final **Q-table** and the **learned policy** (best action for each state).

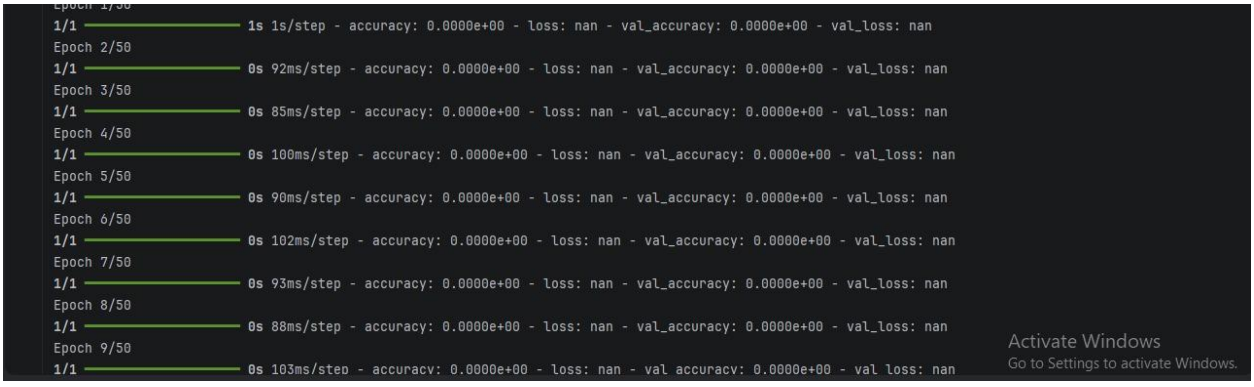
Step 6: Test the trained agent with ε = 0 over several episodes and compute the **success rate** and **average reward**.

Step 7: Vary **α, γ and ε** and **compare** their effect on convergence speed and final performance, then **conclude**.

Program&Result:







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Epoch 17/50
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 107ms/step - accuracy: 0.0000e+00 - loss: nan - val_accuracy: 0.0000e+00 - val_loss: nan
Epoch 18/50
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 89ms/step - accuracy: 0.0000e+00 - loss: nan - val_accuracy: 0.0000e+00 - val_loss: nan
Epoch 19/50
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 93ms/step - accuracy: 0.0000e+00 - loss: nan - val_accuracy: 0.0000e+00 - val_loss: nan
Epoch 20/50
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 111ms/step - accuracy: 0.0000e+00 - loss: nan - val_accuracy: 0.0000e+00 - val_loss: nan
Epoch 21/50
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 99ms/step - accuracy: 0.0000e+00 - loss: nan - val_accuracy: 0.0000e+00 - val_loss: nan
Epoch 22/50
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 98ms/step - accuracy: 0.0000e+00 - loss: nan - val_accuracy: 0.0000e+00 - val_loss: nan
Epoch 23/50
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 88ms/step - accuracy: 0.0000e+00 - loss: nan - val_accuracy: 0.0000e+00 - val_loss: nan
Epoch 24/50
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 97ms/step - accuracy: 0.0000e+00 - loss: nan - val_accuracy: 0.0000e+00 - val_loss: nan
Epoch 25/50
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 88ms/step - accuracy: 0.0000e+00 - loss: nan - val_accuracy: 0.0000e+00 - val_loss: nan
Epoch 26/50
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 87ms/step - accuracy: 0.0000e+00 - loss: nan - val_accuracy: 0.0000e+00 - val_loss: nan
Epoch 27/50
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 103ms/step - accuracy: 0.0000e+00 - loss: nan - val_accuracy: 0.0000e+00 - val_loss: nan
Epoch 28/50
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 89ms/step - accuracy: 0.0000e+00 - loss: nan - val_accuracy: 0.0000e+00 - val_loss: nan
Epoch 29/50
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 89ms/step - accuracy: 0.0000e+00 - loss: nan - val_accuracy: 0.0000e+00 - val_loss: nan
Epoch 30/50
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 98ms/step - accuracy: 0.0000e+00 - loss: nan - val_accuracy: 0.0000e+00 - val_loss: nan
Epoch 31/50
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 87ms/step - accuracy: 0.0000e+00 - loss: nan - val_accuracy: 0.0000e+00 - val_loss: nan
Epoch 32/50
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 99ms/step - accuracy: 0.0000e+00 - loss: nan - val_accuracy: 0.0000e+00 - val_loss: nan
Epoch 33/50
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 84ms/step - accuracy: 0.0000e+00 - loss: nan - val_accuracy: 0.0000e+00 - val_loss: nan
Epoch 34/50
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 85ms/step - accuracy: 0.0000e+00 - loss: nan - val_accuracy: 0.0000e+00 - val_loss: nan
Epoch 35/50
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 103ms/step - accuracy: 0.0000e+00 - loss: nan - val_accuracy: 0.0000e+00 - val_loss: nan
Epoch 36/50
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 87ms/step - accuracy: 0.0000e+00 - loss: nan - val_accuracy: 0.0000e+00 - val_loss: nan
Epoch 37/50
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 86ms/step - accuracy: 0.0000e+00 - loss: nan - val_accuracy: 0.0000e+00 - val_loss: nan
Epoch 38/50
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 100ms/step - accuracy: 0.0000e+00 - loss: nan - val_accuracy: 0.0000e+00 - val_loss: nan
Epoch 39/50
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 90ms/step - accuracy: 0.0000e+00 - loss: nan - val_accuracy: 0.0000e+00 - val_loss: nan
Epoch 40/50
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 99ms/step - accuracy: 0.0000e+00 - loss: nan - val_accuracy: 0.0000e+00 - val_loss: nan
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Epoch 40/50
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 99ms/step - accuracy: 0.0000e+00 - loss: nan - val_accuracy: 0.0000e+00 - val_loss: nan
Epoch 41/50
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 93ms/step - accuracy: 0.0000e+00 - loss: nan - val_accuracy: 0.0000e+00 - val_loss: nan
Epoch 42/50
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 83ms/step - accuracy: 0.0000e+00 - loss: nan - val_accuracy: 0.0000e+00 - val_loss: nan
Epoch 43/50
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 103ms/step - accuracy: 0.0000e+00 - loss: nan - val_accuracy: 0.0000e+00 - val_loss: nan
Epoch 44/50
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 88ms/step - accuracy: 0.0000e+00 - loss: nan - val_accuracy: 0.0000e+00 - val_loss: nan
Epoch 45/50
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 96ms/step - accuracy: 0.0000e+00 - loss: nan - val_accuracy: 0.0000e+00 - val_loss: nan
Epoch 46/50
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 86ms/step - accuracy: 0.0000e+00 - loss: nan - val_accuracy: 0.0000e+00 - val_loss: nan
Epoch 47/50
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 84ms/step - accuracy: 0.0000e+00 - loss: nan - val_accuracy: 0.0000e+00 - val_loss: nan
Epoch 48/50
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 103ms/step - accuracy: 0.0000e+00 - loss: nan - val_accuracy: 0.0000e+00 - val_loss: nan
Epoch 49/50
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 91ms/step - accuracy: 0.0000e+00 - loss: nan - val_accuracy: 0.0000e+00 - val_loss: nan
Epoch 50/50
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 93ms/step - accuracy: 0.0000e+00 - loss: nan - val_accuracy: 0.0000e+00 - val_loss: nan
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 75ms/step
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